INCH-POUND

MIL-PRF-27639D
20 JUNE 1997
SUPERSEDING
MIL-L-27639C (USAF)
18 September 1970

PERFORMANCE SPECIFICATION

VALVE, PNEUMATIC INFLATOR (LIFE PRESERVER, FLU-1A/P)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of pneumatic inflator valve (life preserver, FLU-1A/P).

2. APPLICABLE DOCUMENTS

2.1. General. The documents listed in this section are cited in sections 3 and 4 of this specification. These lists do not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of these lists, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents

2.2.1 Specifications and standards The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the applicable issues of these documents are those listed in the specific issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to: RESOURCES & LOGISTICS SERVICES DIVISION, SA-ALC/TILDD, 485 QUENTIN ROOSEVELT DRIVE BLDG 171, KELLY AFB TX 78241-5916, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A **FSC 4220**

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SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-25369 Cylinders, Carbon Dioxide, For the Inflation of Pneumatic Life Preservers

(Unless otherwise indicated, copies of the above specification are available from the DEFENSE AUTOMATED PRINTED SERVICE (DAPS), BLDG 4-D (DPM-DODSSP), 700 ROBBINS AVENUE, PHILADELPHIA PA 19111-5094)

2.3 Non-Government publications The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the applicable issues of the documents which have been adopted by the DoD are those listed in the specific issue of the DoDISS cited in the solicitation. Unless otherwise specified, the documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4-1993 Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to: THE AMERICAN NATIONAL STANDARDS INSTITUTE, 11 WEST 42ND ST, NEW YORK NY 10036)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-B-117 Standard Test Method of Salt Spray (FOG) Testing

(Application for copies should be addressed to: THE AMERICAN SOCIETY FOR TESTING AND MATERIALS, 1916 RACE STREET, PHILADELPHIA PA 19103-1187)

2.4 Order of precedence In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. If there is a conflict between the contents of this specification and an associated specification or specification sheet, the associated specification or specification sheet should apply.

3. **REQUIREMENTS**

3.1 General. Figure 1 represents a general concept of a pneumatic inflator valve. It should be used for general dimensional purposes such as length, width, and thickness but should not be construed as the only possible design approach.

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3.2 First article. When a first article is specified (see 6.2 and 6.4) it shall be subjected to first article inspection in accordance with 4.5.

3.3 Age. Unless otherwise specifically approved by the procuring activity, all materials and component parts used in the inflator valve shall have been manufactured not more than 12 months prior to the date of delivery to the Government.

3.4 Materials. All materials shall be suitably treated to resist corrosion due to electrolytic decomposition, fungus, salt spray, and any other environmental conditions that may be encountered during operational use or storage. Cadmium and other materials identified as pollutants by the Environmental Protection Agency (EPA) should be avoided where possible.

3.5 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the materials meet or exceed the operational or maintenance requirements, and promote economically advantageous life cycle costs.

3.6 Operation. The pneumatic inflator valve shall be capable of being easily operated in any world-wide environmental conditions that may be encountered by aircrew personnel. The unit shall function easily when activated by personnel wearing all types of flight gear.

3.7 Housing. Housing shall be of any material that is currently accepted in industry for life support items.

3.8 Components.

3.8.1 Inflator valve. The inflator valve discharge mechanism shall consist of an external lever and an internal piercing pin and shall have a seal that will prevent leakage of pressure from the life preserver cell when the lever is in the operated (locked) position. The inlet shall be threaded to accept a carbon dioxide (CO_2) cylinder conforming to MIL-PRF-25369. A locking mechanism shall be provided to retain the CO_2 cylinder in the inlet. An outlet shall be provided to mount the inflator valve on the valve stem of the preserver cell.

3.8.2 Actuating lever. All edges of the lever and the case depth shall have a radius of 0.010 inch. The lever shall be 0.1094 inch thick and shall be held in a stationary position while in operation. In the unoperated position, the lever shall lay against the side of the inflator valve, recessed in the housing.

3.8.3 Piercing pin. The piercing pins shall have a case depth of 0.010 inch and shall be manufactured of a material that is currently recognized for life support items. A seal shall be provided to prevent leakage of the pressure from the life preserver cell after the piercing pin has been retracted.

3.8.3.1 Piercing pin spring. A piercing pin spring shall be provided to retract the pin immediately from the opening of the punctured CO_2 cylinder.

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3.8.4 Outlet. Each inflator valve shall be capable of mounting on an inflator valve stem. The gas passages in the outlet shall align with the holes in the side of the stem regardless of which side of the outlet is toward the base of the stem.

3.8.4.1 Vent passages. Passages shall be provided in the housing for conducting gas from the punctured CO_2 cylinder to the outlet when the lever is in the operated position. Passages shall also be provided for venting residual air that may be trapped in the life preserver cell, from the outlet through the inflator valve, when the lever is in the unoperated position.

3.8.5 Valve stem cap. A valve stem cap shall be provided to hold the inflator valve in place on the valve stem.

3.8.6 Lanyard. A lanyard shall be attached to the free end of the actuating lever.

3.8.7 Gaskets. Gaskets shall be provided to seal the attachment points of the outlet and the inlet of the inflator valve.

3.9 Performance.

3.9.1 Operating force without CO_2 cylinder. The inflator valve shall operate with a load of not more than 2 pounds attached to the lanyard end of the actuating lever.

3.9.2 Piercing mechanism. The piercing mechanism shall be capable, for the life of the inflator valve, of puncturing a CO_2 cylinder conforming to MIL-PRF-25369. The mechanism shall be positioned so as not to accidentally puncture the CO_2 cylinder due to vibrations or other motion encountered in an operational or storage environment.

3.9.3 Lever operation. A stop shall be provided to prevent the lever from traveling a distance of more than 90 degrees from the horizontal axis of the housing. A means shall be provided to return the lever to within 10 degrees from the unoperated position immediately after discharge of the CO_2 cylinder conforming to MIL-PRF-25369.

3.9.4 Hydrostatic pressure. The inflator valve shall satisfactorily withstand a hydrostatic pressure of 1500 pounds per square inch (PSI) without deformation or leakage.

3.9.5 Salt spray. The inflator valve shall operate properly after 96 hours of exposure to a salt spray environment.

3.9.6 Leakage. The inflator valve shall display no evidence of leakage during normal operation.

3.9.7 Operating force with cylinder. With an actuating load of not more than 20 inch-pounds, the inflator valve shall pierce and discharge a CO_2 cylinder conforming to MIL-PRF-25369 in not more than 2.5 seconds.

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3.9.8 Lock. The lock shall be such that a CO_2 cylinder installed with a torque of 60 inch-pounds can be loosened with a torque of not less than 30 or more than 60 inch-pounds.

3.9.9 Weight. The total weight of the pneumatic inflator valve shall not exceed 60 grams.

3.9.10 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of 3.9.11 shall govern the manufacturer's part number and changes thereto.

3.9.11 Markings. The pneumatic inflator valve shall be marked for identification with the following information:

VALVE, PNEUMATIC INFLATOR (LIFE PRESERVER, FLU-1A/P)
Manufacturer's Identification
Contract or Order No.
Date of Manufacture

3.9.11.1 Identification. The identification marking on the inflator valves shall be durable enough to last through the normal life expectancy of the item and be capable of withstanding the environmental tests and cleaning procedures specified. All markings shall be clearly readable. Identification marking on identification plates shall be of a color which is in contrast to the color of the surface of the plate. Identification tag marking, when used, shall be permanent to the extent required for the utilization of the item. Whenever practical, the marking of the item shall be located in such a manner as to allow it being visible during normal operational use. The use of marking materials that create environmentally hazardous conditions is prohibited. The marking of items shall be accomplished in a manner which shall not adversely affect the life and utility of the item (see 6.5).

3.9.12 Safety. The pneumatic inflator valve shall be functional without sharp edges, burrs, or protrusions that might cause injury or prevent the use of the item.

3.9.13 Workmanship. The inflator valve, including all parts and equipment, shall be constructed in accordance with commonly accepted industrial workmanship standards.

3.9.14 Actuator safety restraint. A replaceable actuator safety restraint shall be provided to prevent the actuator from operating accidentally and causing the piercing mechanism to puncture the CO_2 cylinder.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3)

4.2 First article inspection. First article inspection shall consist of all examinations and tests described in 4.7.

4.3 Conformance inspection. This inspection shall include the tests in 4.3.1 and 4.3.2.

4.3.1 Individual inspection. Individual inspections shall consist of the following tests as described under 4.7.

a. Examination of product (see 4.7.1)
b. Operation test without CO_2 cylinder (see 4.7.2)
c. Piercing mechanism operation (see 4.7.3)
d. Actuator operation (see 4.7.4)

4.3.2 Sampling inspection.

4.3.1 Lot size. A lot shall consist of inflator valves manufactured under essentially the same conditions and submitted for inspection at the same time. The lot size for sampling plans A and B shall consist of up to 500 units. The lot size for sampling plan C shall be in accordance with ANSI/ASQC Z1.4-1993 inspection level S-3. Sampling plans A, B, and C shall be conducted in accordance with the quality assurance section of the contract. Non-conformance requirements for sampling plans A, B, and C shall be provided in the quality assurance section of the contract (see 6.7).

4.3.1.1 Sampling plan A. One inflator valve shall be selected at random from the first lot, and from each fourth lot thereafter, and shall be subjected to the following tests as described under 4.7.

a. Hydrostatic pressure (see 4.7.5)
b. Salt spray (see 4.7.6)
c. Safety actuator restraint (see 4.7.10)

4.3.1.2 Sampling plan B. Ten inflator valves shall be selected at random from each lot and shall be subjected to the following tests as described under 4.7.

a. Leakage (see 4.7.7)
b. Operation test with CO_2 cylinder (see 4.7.8)
c. Lock (see 4.7.9)

4.3.1.3 Sampling plan C. Upon completion of the individual tests specified in 4.2, random samples shall be selected from each lot in accordance with inspection level S-3 of the

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ANSI/ASQC Z1.4-1993. The sample size shall be based only on the applicable size code letter corresponding to inspection level S-3. Each inflator valve selected as a sample unit shall be forwarded to an inspection activity (see 6.3) selected by the procuring activity for the following tests as described under 4.7.

a. Examination of product (see 4.7.1)
b. Operation test without CO_2 cylinder (see 4.7.2)
c. Piercing mechanism operation (see 4.7.3)
d. Actuator operation (see 4.7.4)

4.4 Inspection conditions.

4.4.1 Test items. Unless otherwise indicated in the tests methods, all performance, environmental and test requirements are based on a pneumatic inflation valve complete with replacable components.

4.4.2 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury, a temperature of 70 ± 10 degrees Fahrenheit, and a relative humidity of 80 percent or less. If these conditions are not obtainable, proper corrections shall be made to the tests or the test results to compensate for the different atmospheric conditions.

4.5 First article inspection. If first articles are specified (see 6.2 and 6.4), the first article shall consist of two units. First article inspection may consist of one or more of the described sampling plans (see 4.3).

4.6 Verification alternatives. The pneumatic inflator valve shall meet all performance requirements in section 3. Verification methods indicated in this section are mandated after contract execution. Alternative test methods, techniques, or equipment, including the application of statistical process control or cost effective sampling procedures, can be proposed and used provided that these alternatives offer equal assurance that all product requirements are met. Such acceptable alternative verification approaches shall be inserted into the contract as negotiated replacements. In case of dispute as to whether any alternatives provide necessary equal assurance, the verification provisions as specified apply.

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4.7 Methods of inspection. Table 1 provides a cross-reference matrix of the requirements tested or verified in the paragraphs below.

TABLE 1: Requirements cross-reference matrix.

REQUIREMENT	VERIFICATION
3.9.1	4.7.2
3.9.2	4.7.3
3.9.3	4.7.4
3.9.4	4.7.5
3.9.5	4.7.6
3.9.6	4.7.7
3.9.7	4.7.8
3.9.8	4.7.9
3.9.9	4.7.1
3.9.10	4.7.1
3.9.11	4.7.1
3.9.12	4.7.1
3.9.13	4.7.1
3.9.14	4.7.10

4.7.1 Examination of product. Each inflator valve shall be visually examined to determine compliance with the requirements of this specification with respect to workmanship, materials, markings, dimensions, weight, and interchangeability as cited in section 3.

4.7.2 Operation test without CO_2 cylinder. The inflator valve shall be held in an inverted position with a 2-pound load attached to the actuator with the actuator held in the normal position. The load shall be released and the actuator shall operate the piercing mechanism. The actuator safety restraint shall not be used to secure the actuator during this test.

4.7.3 Piercing mechanism operation. The piercing mechanism shall be operated through its full range of motion. The mechanism shall not be capable of piercing the CO_2 cylinder while in the stored position when it is subjected to a vibration frequency specified in the procuring document (see 6.2).

4.7.4 Actuator operation. The actuating lever of each inflator valve shall be operated through an arc of not more than 90 degrees. The piercing mechanism shall pierce the installed CO_2 cylinder and then retract to its original position with a travel of the actuator through an arc of not more than 90 degrees in line with the longitudinal axis of the inflator valve.

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4.7.5 Hydrostatic pressure. A discharged CO_2 cylinder shall be installed for this test. A hydrostatic pressure of 1500 PSI shall be applied to the outlet of the inflator valve for a period of not less than 5 minutes. The actuator shall be in the operated position. There shall not be any deformation or leakage of the inflator valve. After completion of this test, the inflator valve shall be subjected to the tests specified in 4.7.6.

4.7.6 Salt spray. The inflator valve shall be subjected to a 96-hour salt spray test in accordance with the test methods in ASTM-B-117. After the salt-spray test, there shall be no corrosion. After the completion of the salt-spray test, the inflator valve with a charged CO_2 cylinder conforming to MIL-PRF-25369 installed, shall be held in an inverted position with a 20-pound load attached to the end of the actuator and with the actuator held in the normal position. When the load is released, the actuator shall operate the piercing mechanism which shall pierce and discharge the CO_2 cylinder in not more than 2.5 seconds. The actuator safety restraint shall not be used to secure the actuator during this test.

4.7.7 Leakage. With the inflator valve mounted on a valve stem, with the valve stem cap in place , the actuator shall be moved slowly through a normal cycle of operation. A pressure of 40 PSI shall be applied to the outlet and any indication of leakage shall be cause for rejection.

4.7.8 Operation test with CO_2 cylinder. The inflator valve shall be mounted on a valve stem with the valve stem cap in place. A charged CO_2 cylinder conforming to MIL-PRF-25369 shall be installed. The inflator assembly shall be held in an inverted position with a load attached to the lever of the actuator and the actuator held in the normal position. The outlet shall be equipped with a chamber that shall provide between 1.5 and 2.5 PSI back pressure on the inflator valve. The load shall be such that a force of 20 pounds is applied to the actuator. When the load is released, the actuator shall operate the piercing mechanism which shall pierce and discharge the CO_2 cylinder in not more than two and one half seconds. The actuator safety restraint shall not be used to secure the actuator in this test. No leakage shall be evident during the duration of this test.

4.7.9 Lock. The lock on the installed CO_2 cylinder (conforming to MIL-PRF-25369) shall be tested by applying a torque of 60 inch-pounds in a counterclockwise direction to loosen the installed CO_2 cylinder. The CO_2 cylinder shall release with a torque of not less than 30 nor more than 60 inch-pounds.

4.7.10 Actuator safety restraint. The inflator valve shall be held in an inverted position with a 20-pound load attached to the actuator. The actuator shall be held in the normal position with the actuator safety restraint installed. When the load is released , the replacable, actuator safety restraint shall prevent the actuator from operating the piercing mechanism during the test.

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5. PACKAGING

5.1 General. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point packaging activity within the Military Department or Defense Agency or within the Military Department's System. Packaging data retrieval is available from the managing Military Department or Defense Agency automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The pneumatic inflator valve covered by this specification is intended to provide a means of piercing a CO_2 cylinder and transferring the gas into a life preserver. It is used as a component of the LPU-3/P and the LPU-10/P life preservers.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of the specification.
b. Issue of the DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).
c. Packaging requirements (see 5.1).
d. When first article is required (see 3.2, 4.2).
e. Type of CO_2 cylinder to be used. (see 3.8.1)
f. Conformance requirements for sampling plans A, B, and C. (4.3.1, 4.3.2)

6.3 Test samples for sampling plan C. The contracting officer should identify the responsible inspection activity for the specific acquisition and should include specific instructions in the acquisition documents for use by the inspection activity. The test samples rejected under sampling plan C should be returned to the manufacturer.

6.4 First article. When a first article inspection is required, the item(s) should be a first article sample. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results, and disposition of first articles. Invitations for bids should state that the Government reserves the right to waive the requirements for samples for the first article inspection from those bidders offering a product which has been previously acquired or tested by the government and that bidders offering such products, who wish to rely on such production or test, should furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.5 Marking methods. The tables in MIL-STD-130, “Marking Methods” and “Criteria in Selection of Marking Methods”, provide guidance for identification marking methods, and are not mandatory.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.7 Quality assurance provisions. Acquisition documents should provide quality assurance provisions designating which sampling plans are to be used. Acceptable quality levels for sampling plans A and B should also be provided.

6.8 Defects in inflating assemblies already accepted. Procuring documents should specify recall or corrective procedures if an investigation of a test failure indicates that defects may exist in inflating assemblies already accepted. The contractual documents should require the contractor to advise the procuring activity of all the defects likely to be found and the method of correcting them. If further inspection should disclose defects which would render the item unusable or unsafe, the supplier should be responsible for recalling those lots identified as containing unsafe items or, if field repairable, should provide materials and labor as necessary to correct the identified deficiencies.

6.9 Key Words.
Life preserver
Piercing pin
CO_2 cylinder
Inflator valve
Piercing pin spring
Actuating lever
Valve stem cap

Custodian:
Air Force - 99

Reviewers:
DLA - CS
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Preparing Activity:
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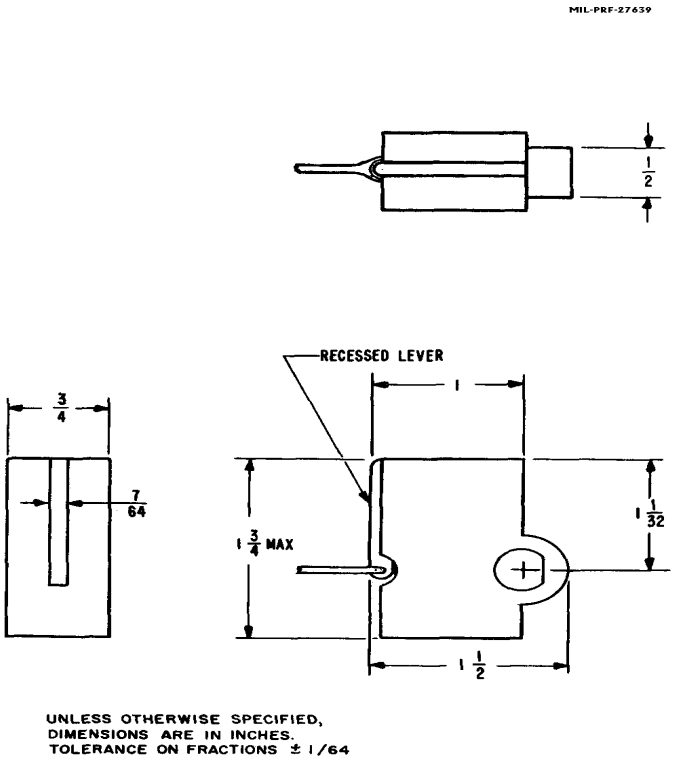

*FIGURE 1. Inflator

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